Technical Documentation

Project Overview

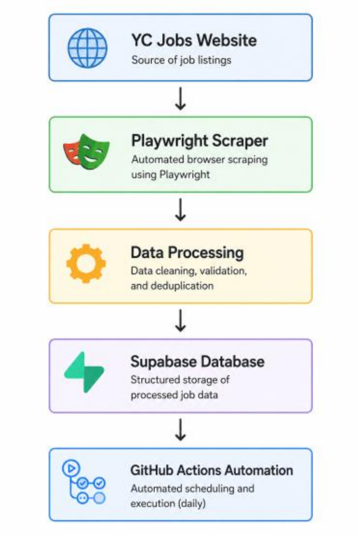
The Automated Data Collection System is designed to automatically extract job listings from the YC Jobs platform, process the collected information, store it in a database, and execute periodically without manual intervention.

The solution utilizes modern web automation technologies and cloud-based infrastructure.

System Architecture

The system consists of the following components:

1. YC Jobs Website
2. Playwright Scraper
3. Data Cleaning and Deduplication
4. Supabase Database
5. GitHub Actions Scheduler



Technology Stack

Programming Language

JavaScript (Node.js)

Reason:

- Efficient asynchronous operations
- Large ecosystem
- Excellent support for web automation

Browser Automation Framework

Playwright

Reason:

- Supports modern web applications
- Handles dynamic content
- Reliable page interaction
- Cross-browser compatibility

Database

Supabase

Reason:

- Cloud-hosted PostgreSQL
- Easy integration
- RESTful APIs
- Real-time dashboard

Automation Platform

GitHub Actions

Reason:

- Free automation workflows
- Scheduled execution support
- Easy repository integration

Data Collection Process

Step 1: Browser Initialization

The Playwright script launches a browser instance and navigates to the YC Jobs website.

Step 2: Page Loading

The scraper waits until all required content is fully rendered.

Step 3: Data Extraction

Relevant elements are identified using CSS selectors.

Collected fields include:

- Company Name
- Job Title
- Job URL
- Location

Step 4: Data Storage

The extracted records are temporarily stored in JSON format.

Example:

{
"company":"Example Startup",

"title":"Software Engineer",
"url":"https://example.com/job"
}

Data Processing

Deduplication

Duplicate job records are removed before database insertion.

Benefits:

- Reduced storage usage
- Improved data quality
- Easier analysis

Validation

Records are checked to ensure:

- Required fields exist
- URLs are valid
- Empty records are removed

Database Design

Database Platform

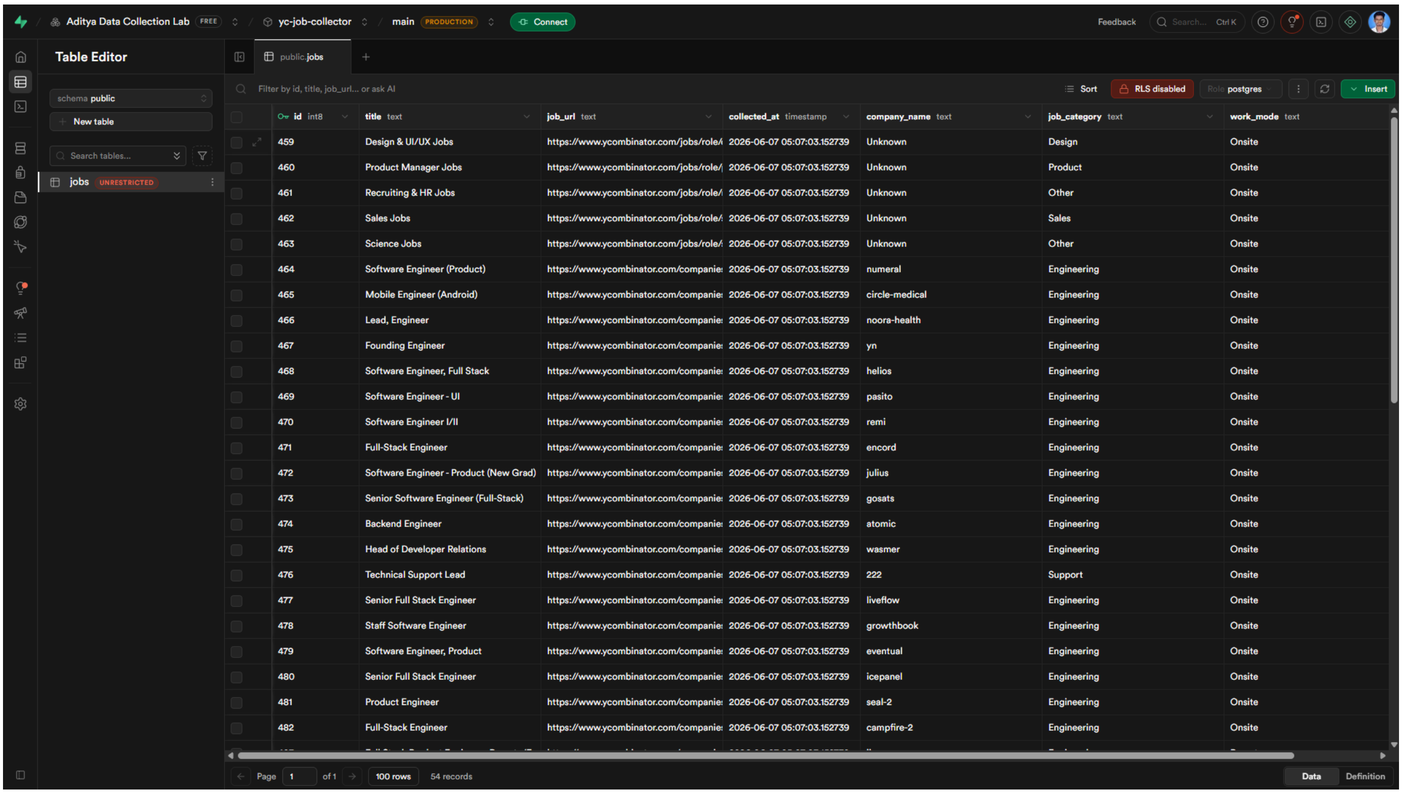
Supabase PostgreSQL

Table: jobs

Fields:

Column	Type
id	Integer
company	Text
job_title	Text
job_url	Text
location	Text
collected_at	Timestamp

The database stores all processed job records for future analysis.



id int8	title text	job_url text	collected_at timestamp	company_name text	job_category text	work_mode text
459	Design & UI/UX Jobs	https://www.ycombinator.com/jobs/role/	2026-06-07 05:07:03.152739	Unknown	Design	Onsite
460	Product Manager Jobs	https://www.ycombinator.com/jobs/role/	2026-06-07 05:07:03.152739	Unknown	Product	Onsite
461	Recruiting & HR Jobs	https://www.ycombinator.com/jobs/role/	2026-06-07 05:07:03.152739	Unknown	Other	Onsite
462	Sales Jobs	https://www.ycombinator.com/jobs/role/	2026-06-07 05:07:03.152739	Unknown	Sales	Onsite
463	Science Jobs	https://www.ycombinator.com/jobs/role/	2026-06-07 05:07:03.152739	Unknown	Other	Onsite
464	Software Engineer (Product)	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	numeral	Engineering	Onsite
465	Mobile Engineer (Android)	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	circle-medical	Engineering	Onsite
466	Lead, Engineer	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	noora-health	Engineering	Onsite
467	Founding Engineer	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	yn	Engineering	Onsite
468	Software Engineer, Full Stack	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	helios	Engineering	Onsite
469	Software Engineer - UI	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	pasito	Engineering	Onsite
470	Software Engineer I/II	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	remi	Engineering	Onsite
471	Full-Stack Engineer	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	encord	Engineering	Onsite
472	Software Engineer - Product (New Grad)	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	julius	Engineering	Onsite
473	Senior Software Engineer (Full-Stack)	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	gosats	Engineering	Onsite
474	Backend Engineer	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	atomic	Engineering	Onsite
475	Head of Developer Relations	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	wasmer	Engineering	Onsite
476	Technical Support Lead	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	222	Support	Onsite
477	Senior Full Stack Engineer	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	liveflow	Engineering	Onsite
478	Staff Software Engineer	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	growthbook	Engineering	Onsite
479	Software Engineer, Product	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	eventual	Engineering	Onsite
480	Senior Full Stack Engineer	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	icepanel	Engineering	Onsite
481	Product Engineer	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	seal-2	Engineering	Onsite
482	Full-Stack Engineer	https://www.ycombinator.com/companie	2026-06-07 05:07:03.152739	campfire-2	Engineering	Onsite

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Automation Implementation

GitHub Repository

The project source code is hosted in GitHub.

GitHub Actions Workflow

A workflow file was created to automate execution.

Tasks performed:

1. Install dependencies
2. Execute scraper
3. Process collected data
4. Upload records to Supabase

Benefits

- Fully automated execution
- No manual intervention required
- Consistent data collection
- Easy maintenance

Error Handling

The system includes basic error handling mechanisms.

Examples:

- Network failures
- Page loading issues
- Missing elements
- Database insertion failures

Errors are logged to assist troubleshooting.

Testing and Validation

The system was tested using multiple execution cycles.

Validation checks included:

- Successful page navigation
- Accurate data extraction
- Duplicate removal
- Successful database insertion
- Automated workflow execution

Results confirmed reliable system performance.

Conclusion

The Automated Data Collection System successfully demonstrates end-to-end data acquisition, processing, storage, and automation.

The project achieves all required objectives:

- Automated data collection
- Structured data storage
- Cloud database integration
- Scheduled execution
- Minimal manual intervention

The system provides a scalable foundation for future recruitment analytics and data engineering applications.